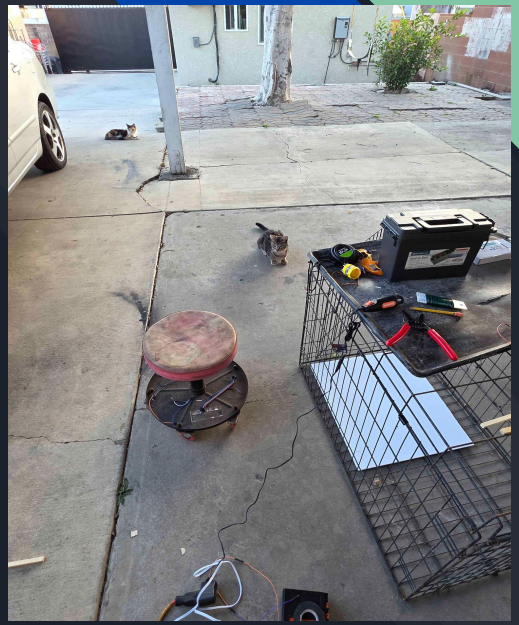
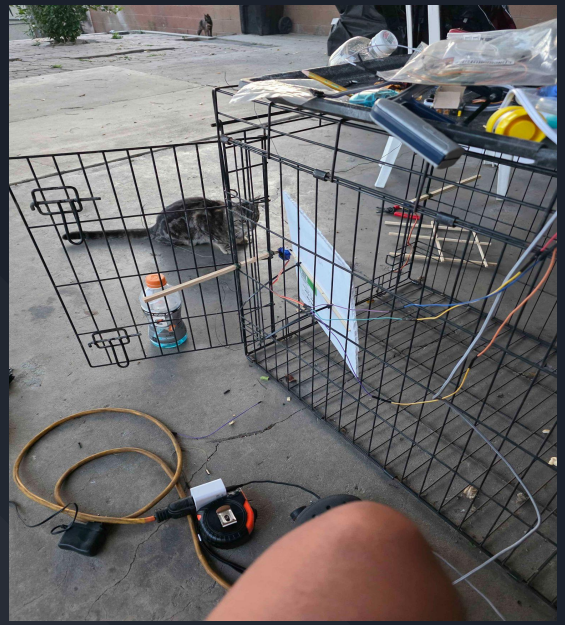
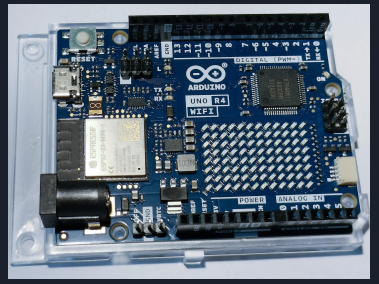
Kennel Bait

By: Jonathan Penaloza & Katherine Poz

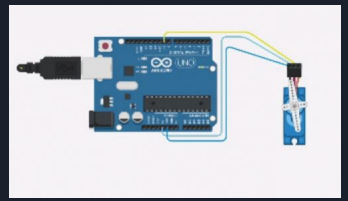




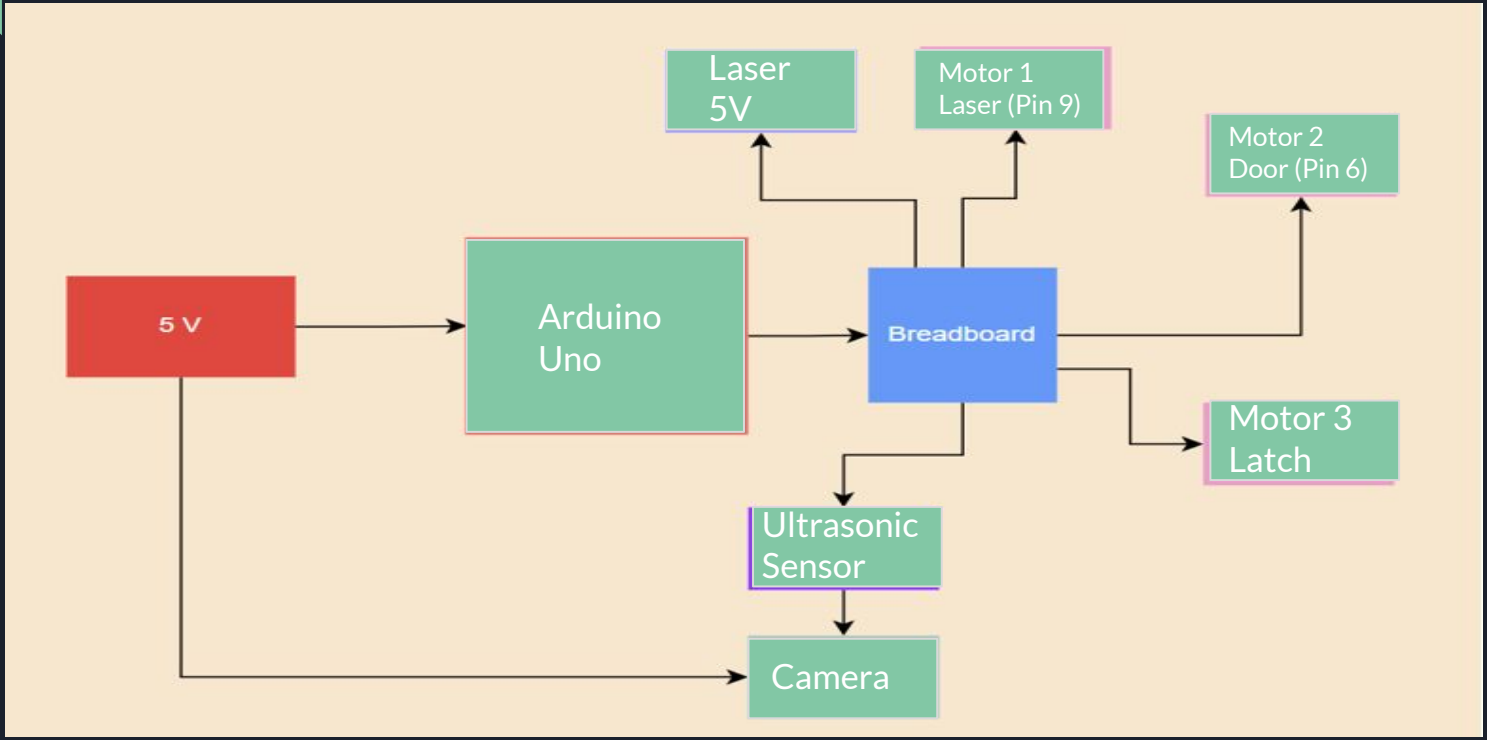
Project Overview



- The main purpose of this project is to capture stray kittens and cats
- We wanted to demonstrate how engineering can be applied to the real world
- The device we used in this project is an arduino uno
- The components being used are servos, a camera, and a ultrasonic sensor



Project Overview



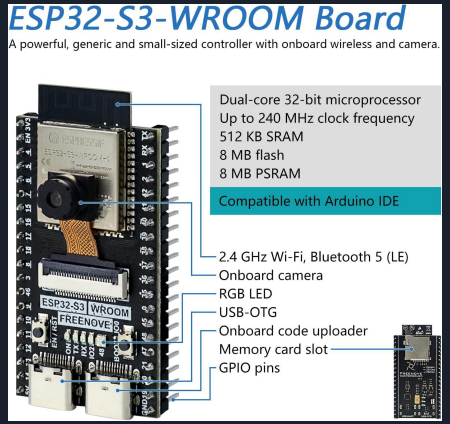
System Architecture

The overall system architecture was designed using:

- Arduino Uno: Microcontroller for simplicity, reliability, and ease of hardware integration. It is our primary microcontroller which provides GPIO pins, PWM outputs and accessible programming environment
- Servo Motor 1: The first subsystem is used for the laser pointer which will swing from 90 degrees to 0 degrees
- Sensor: Once the servo motor 1 reaches 0 degrees the sensor will recognize the object and will communicate with servo motor 2.
- Servo Motor 2: The second subsystem is used for the door which will close once the kitten enters
- Camera: The camera will record manually

Interfaces and Peripheral Used

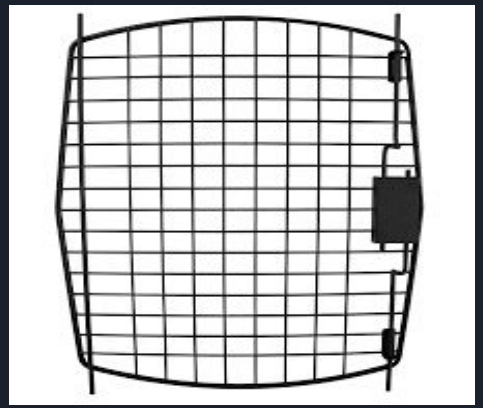
- GPIO: Ultrasonic Sensor
- PWM : Servos
- Ethernet: Camera

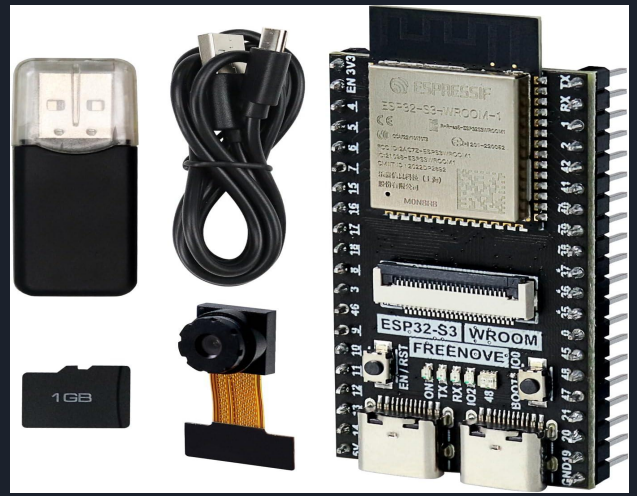


Servo Motor 1	Servo Motor 2	Ultrasonic Sensor
Laser	Door	—
Pin 9	Pin 6	Pin 2 Transmitter (TX), Pin 4 Receiver(RX)

Verification and Testing

- Door
- Laser
- Camera





Project Demo

Project Demo

Thank You